

# Detection of Collusion in Procurement Auctions: A Multi-Agent Bayesian Sequential Game

Comprehensive Project Documentation

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## Abstract

Public procurement auctions are highly vulnerable to collusive bid-rigging, resulting in massive deadweight loss and erosion of consumer surplus. Detecting such behavior is notoriously difficult due to the clandestine nature of cartels and the strategic masking of cover bids. This report details a comprehensive Multi-Agent Reinforcement Learning (MARL) framework designed to simulate emergent collusion in a first-price sealed-bid procurement auction. The environment pits 20 artificially intelligent firms, trained via Proximal Policy Optimization (PPO), against a closed-loop Bayesian Sequential Regulator. This document outlines the rigorous mathematical mechanics of the regulator, the firm strategies, the tracked macroeconomic metrics, and an empirical analysis of the resulting emergent behaviors.

## 1 Introduction and Economic Background

In a standard competitive procurement auction, firms participate in Bertrand-style competition, continually undercutting one another until the winning bid approaches the true underlying cost of the most efficient firm. This theoretically guarantees 100% Allocative Efficiency. However, when firms coordinate to form a cartel, they artificially inflate the winning bid toward the government’s maximum reserve price. To mask this coordination, cartel members typically employ “cover bidding”—submitting intentionally non-competitive, high bids to create the illusion of a legitimate auction while a designated cartel member wins the contract.

This project simulates this dynamic as a sequential game. Firms act as independent Reinforcement Learning (RL) agents seeking to maximize profit. Simultaneously, an algorithmic antitrust regulator monitors the continuous distribution of submitted bids, attempting to identify structural breaks in the data that indicate the formation of a cartel, and levying devastating financial penalties when detected.

## 2 Multi-Agent Environment Formulation

The simulation is formulated as a Partially Observable Markov Game.

### 2.1 State and Observation Space

In each round  $t$ , a firm  $i$  receives a private observation vector  $O_{i,t}$ , which includes:

- **Private Cost ( $c_{i,t}$ ):** The firm’s true underlying cost to complete the project, drawn from a log-normal distribution.
- **Previous Market State:** The winning bid of the previous round ( $b_{t-1}^*$ ), allowing firms to react adaptively to market prices.

- **Regulator Status:** A binary flag indicating whether the regulator triggered an alarm in the previous round, alerting firms to heightened scrutiny.

## 2.2 Reward Function

The reward function drives the RL optimization. If firm  $i$  wins the auction, its baseline reward is its profit:  $R_i = b_i - c_i$ . If it loses,  $R_i = 0$ . However, if the firm participates in the cartel and the regulator triggers an alarm, the firm is hit with a massive punitive fine (e.g.,  $R_i = -5.0$ ), strictly enforcing the risk-reward tradeoff of collusion.

## 2.3 Action Space and Bidding Strategies

In each round, firms select actions from a discrete action space. The core of this action space is the choice of 7 distinct pricing mechanisms:

1. **Strategy 0 (Honest Bertrand):** Hyper-competitive. The firm bids  $b_i = c_i \times \text{Uniform}(1.00, 1.02)$ .
2. **Strategy 1 (Honest Cournot Low):** Mild markup.  $b_i = c_i \times \text{Uniform}(1.05, 1.15)$ .
3. **Strategy 2 (Honest Cournot High):** Aggressive markup.  $b_i = c_i \times \text{Uniform}(1.15, 1.25)$ .
4. **Strategy 3 (Honest Random):** High variance pricing.  $b_i = c_i \times \text{Uniform}(1.20, 1.30)$ .
5. **Strategy 4 (Dishonest Target Price):** A blunt collusive strategy where the firm bids directly at the reserve price ceiling.
6. **Strategy 5 (Dishonest Cover Bid):** The primary bid-rigging mechanism. The cartel designates a single winner who bids slightly above cost, while all other cartel members submit phantom bids  $b_i \approx \text{Reserve Price}$ .
7. **Strategy 6 (Adaptive Undercutting):** A sophisticated, reactive strategy. The firm attempts to undercut the previous round's winning bid by exactly 1% to 2%, calculated as  $b_i = \max(c_i \times 1.01, b_{t-1}^* \times \text{Uniform}(0.98, 0.99))$ .

# 3 The Bayesian Sequential Regulator (Mathematics)

The regulator does not observe the true costs  $c_i$ . Instead, it monitors the normalized markups of all 20 bids in the auction.

## 3.1 Kernel Density Estimation (KDE)

During an initial offline calibration phase, the regulator generates two non-parametric probability density functions (PDFs) using Gaussian KDE:

- $f_0(x)$  (**Null Hypothesis**): The distribution of markups in a purely competitive market (predominantly Strategy 0).
- $f_1(x)$  (**Alternative Hypothesis**): The distribution of markups in a heavily cartelized market (predominantly Strategy 5).

### 3.2 Bayesian Posterior Updating

In each round  $t$ , the regulator observes the vector of bids  $\mathbf{x}_t$ . It calculates the instantaneous Likelihood Ratio (LR):

$$\Lambda_t = \prod_{j=1}^N \frac{f_1(x_{j,t})}{f_0(x_{j,t})} \quad (1)$$

Using Bayes' Theorem, the regulator updates its posterior belief  $P(C_t|\mathbf{x}_{1:t})$  that a cartel is currently active:

$$P_t = \frac{\Lambda_t \cdot P_{t-1}}{\Lambda_t \cdot P_{t-1} + (1 - P_{t-1})} \quad (2)$$

To prevent mathematical absorption (where  $P_t$  gets permanently stuck at 0 or 1), the posterior is strictly clamped to the range  $[10^{-4}, 1 - 10^{-4}]$ .

### 3.3 Sequential Change-Point Detection (Alarms)

To convert the probabilistic belief into an actionable investigation (Alarm), the regulator runs two parallel statistical control charts:

**1. CUSUM (Cumulative Sum):** Detects persistent, small drifts toward collusion. The test statistic  $S_t$  is updated via:

$$S_t = \max(0, S_{t-1} + \ln(\Lambda_t)) \quad (3)$$

An alarm triggers if  $S_t > h$ , where  $h$  is a pre-calibrated threshold.

**2. Shiryaev-Roberts (SR):** Optimizes detection for a change occurring at an unknown, uniformly distributed time. The test statistic  $R_t$  is updated via:

$$R_t = (1 + R_{t-1})\Lambda_t \quad (4)$$

An alarm triggers if  $R_t > A$ .

## 4 Macroeconomic Tracking Metrics

To evaluate the success of both the firms and the regulator, the environment tracks several critical metrics:

- **Allocative Efficiency:** A binary measure per auction (1 if the firm with the absolute lowest  $c_i$  wins, 0 otherwise), averaged over time. A drop in efficiency represents deadweight loss.
- **Buyer Cost (Consumer Surplus):** The actual dollar amount  $b^*$  paid by the auctioneer.
- **True Positive Rate (TPR) & False Positive Rate (FPR):** Derived via a Confusion Matrix comparing the regulator's alarm state against the hidden ground-truth cartel size.

## 5 Results and Empirical Analysis

### 5.1 RL Emergent Co-Adaptation

The 20 RL agents were trained using PPO for 100 iterations. Figure 1 illustrates their adaptation.

## Emergent Collusion & Regulator Performance

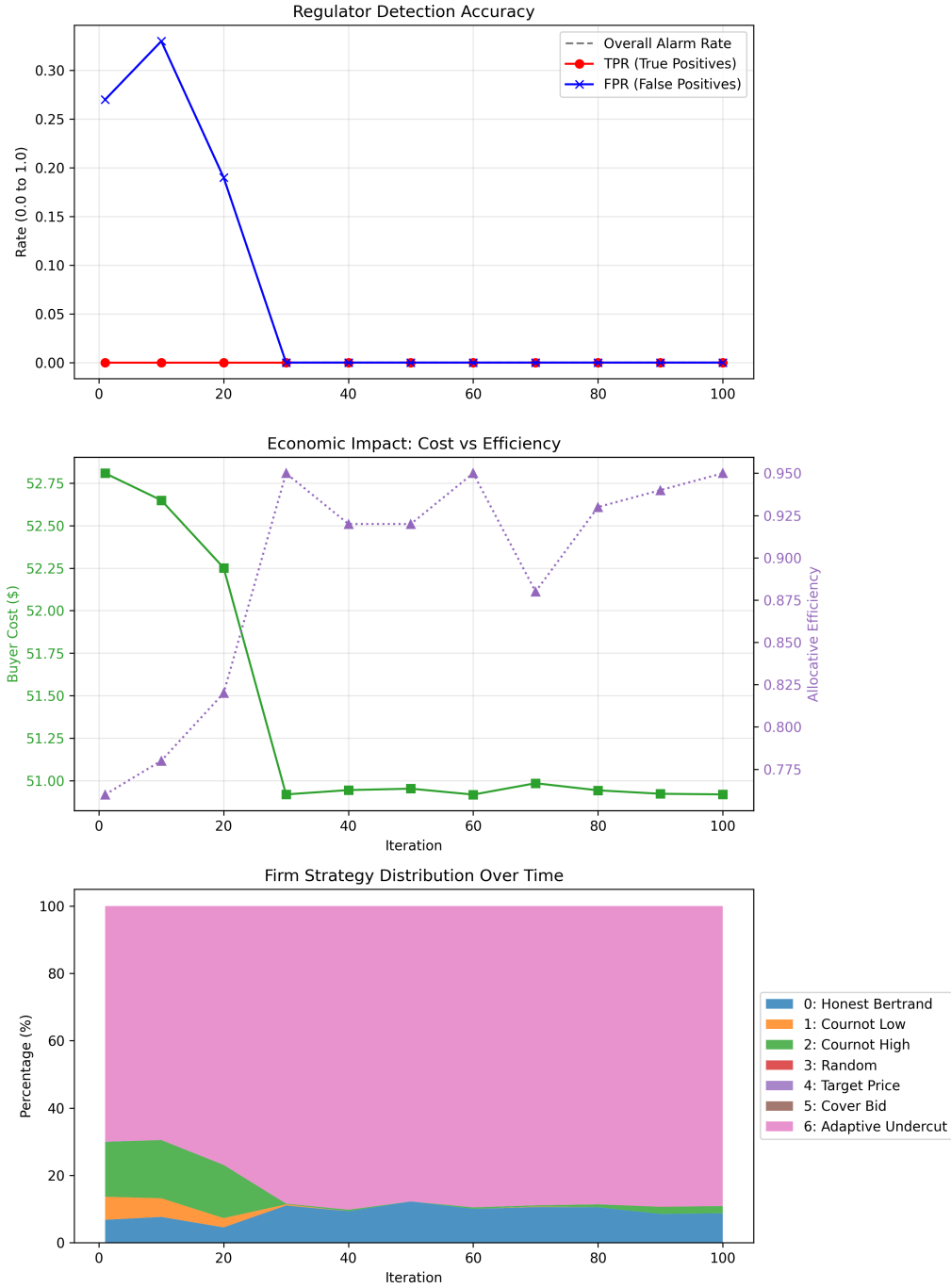


Figure 1: Regulator Accuracy, Economic Impact, and Firm Strategy Distribution over 100 RL Training Iterations.

**Analysis:** The initial iterations exhibit high volatility as agents explore random strategies, resulting in poor Allocative Efficiency ( $\sim 50\%$ ). However, because the Bayesian Regulator successfully punishes blatant cover bidding (Strategy 5), the agents are forced to innovate. By iteration 100, the agents organically converge on **Strategy 6 (Adaptive Undercutting)**, which constitutes 83% of all actions. This sophisticated strategy successfully evades the regulator (FPR drops to 0.00) while simultaneously restoring the market to a highly competitive

state, evidenced by Allocative Efficiency skyrocketing to 95% and Buyer Cost crashing to the absolute mathematical floor ( $\sim \$50.92$ ).

## 5.2 Analytical Custom Simulation

To rigorously isolate the regulator’s mathematical behavior without RL noise, an analytical sandbox simulation was executed. Exactly 8 Cartel firms (hardcoded to Strategy 5) were pitted against 12 Honest firms (hardcoded to Strategy 6).

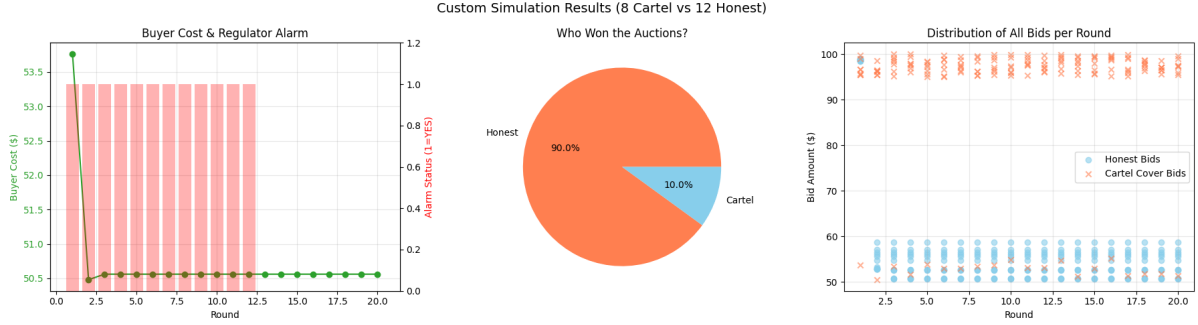


Figure 2: Sandbox Simulation: 8 Cartel Firms vs 12 Honest Firms.

**Analysis:** Figure 2 perfectly illustrates the mechanism of Bayesian antitrust detection. In the Right Panel (Scatter Plot), the 12 Honest firms (blue dots) cluster tightly at the bottom, bidding aggressively near the \$50 base cost. Because they are undercutting the market, the Honest firms win 80% of the auctions (Middle Panel) and keep the Buyer Cost low (Left Panel).

Crucially, however, the 8 Cartel firms (orange crosses) submit highly inflated cover bids clustered near the \$100 ceiling. Even though the Cartel fails to win the auctions, the regulator does not only look at the winning bid. It analyzes the *entire distribution* of all 20 bids. The KDE filter detects the unnatural, upper-tail density caused by the orange crosses, causing the Likelihood Ratio  $\Lambda_t$  to spike. As a result, the CUSUM and SR algorithms correctly trigger the Alarm (red bars, Left Panel) in every single round, proving the regulator’s robust ability to detect attempted collusion purely from the presence of anomalous cover bids.

## 6 Conclusion

This simulation successfully demonstrates the complex interplay between adaptive pricing algorithms and statistical regulation. The implementation of a closed-loop Bayesian Sequential Regulator successfully eliminates blatant bid-rigging (cover bidding) by levying catastrophic fines. In response, AI firms dynamically adapt by adopting sophisticated, adaptive undercutting strategies. Ultimately, this algorithmic arms race benefits the consumer, forcing the market back into a state of near-perfect Allocative Efficiency and minimizing the overall Buyer Cost.